HW 1

Time spent: 3 hours

For an N-point DFT, prove that the basis vectors $v_i, i = 0,1, \ldots, N-1$ satisfy $v_i^H v_j = 0$ for $i \neq j$ and $v_i^H v_i = N$. In other words, the basis vectors are orthogonal and have Euclidean length $\sqrt{N}$.

The basis vectors v_k of the DFT are $e^{-j\frac{2\pi}{N}nk}$

We can check the first relation. Instead of i and j I will use the variables n and m.

We define the inner product as the sum from $k=0$ to $k=N$.

$$
\begin{aligned}
v_n^\dagger v_m = & \\
&= \sum_{k=0}^{N-1} \left(e^{-j\frac{2\pi}{N}kn}\right)^\dagger e^{-j\frac{2\pi}{N}km} \\
&= \sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}k(m-n)}
\end{aligned}
$$

This is a geometric series
$r = e^{\left(j\frac{2\pi}{N}(n-m)\right)}$ and $a = 1$.
This converges to

$$
\begin{aligned}
&\frac{1-r^N}{1-r} \\
&= \left(\frac{1-e^{j\frac{2\pi}{N}(n-m)^N}}{1-e^{j\frac{2\pi}{N}(n-m)}}\right) \\
&= \left(\frac{1-e^{j2\pi(n-m)}}{1-e^{j\frac{2\pi}{N}(n-m)}}\right)
\end{aligned}
$$

For the case $m \neq n$, The numerator is $(1-e^{j2\pi(n-m)})$. N and M are integers, so this is equivalent to $1-e^{j2\pi} = 0$.

if $n = m$, then $r = e^{j2\pi(0)} = 1$. The Geometric series for $r = 1$ is $\sum_{k=0}^{N-1} 1 = N$.

Because the inner product is the $\text{Euclidian length}^2$, the length of the basis vector $v_i = \sqrt{N}$.

This is the first way I did it, which was convoluted but I think technically right. I realized when rereading wikipedia that the formula was different for $r = 1$, which should have been obvious. I'm still proud of this below thing though, so I left it in.
We can also see this as the result from analytic continuation (if we pretend that M and N are real instead of just integers). For the case $m \to n$, the series approaches $\frac{0}{0}$. We can therefore use L'hopitals rule. This yields N, since the derivatives of the top and the bottom are the same (so dividing gets 1) except for the overall factor of N on the numerator.

Prove that if $x[n]$ is purely real, then its N-point DFT has the property $X[k] = X^*[N-k]$ for $k = 0, 1, \ldots, N-1$.

Intuitively:

For a real, double sided DFT indexed from highest negative frequency to zero to highest positive frequency, there is a reflection symmetry where X(highest positive frequency) = X(highest negative frequency), where the $|\text{ highest freq }| = |\text{ highest negative freq }|$.In index notation, with 0 at the highest negative frequency, this is $X(k) = X(N-k)$

Assuming that $x[n]$ is real, we can check the equality by direct substitution

$$X[k] \stackrel{?}{=} X^*[N-k]$$

$$
\begin{aligned}
\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}k_n} &\stackrel{?}{=} \left(\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}(N-k_n)}\right)^* \\
\text{if x is real, then } x[n]^* &= x[n] \\
e^{-j\frac{2\pi}{N}k_n} &\stackrel{?}{=} e^{-j\frac{2\pi}{N}(k_n-N)} \\
&\uparrow e^{-j\frac{2\pi}{N}N}e^{\frac{2\pi}{N}k_n} \\
1 &= e^{-j2\pi} \\
1 &= 1
\end{aligned}
$$

Therefore, if $x[n]$ is real, then $X[k] = X^*[N-k]$.

For $N = 2$ and $N = 3$,

a. Determine the DFT matrix
b. Determine the iDFT matrix
c. Verify that the iDFT matrix is the inverse of the DFT matrix.

We have the generic DFT matrix

$$
\begin{bmatrix}
e^{-j\frac{2\pi}{N}} & e^{-j\frac{2\pi}{N}} & \dots & e^{-j\frac{2\pi}{N}(N-1)} \\
e^{-j\frac{2\pi}{N}\cdot 0} & e^{-j\frac{2\pi}{N}\cdot 1} & \dots & e^{-j\frac{2\pi}{N}(N-1)} \\
\vdots & & & \\
e^{-j\frac{2\pi}{N}} & e^{-j\frac{2\pi}{N}1\cdot(N-1)} & \dots & e^{-j\frac{2\pi}{N}(N-1)^2}
\end{bmatrix}
$$

The generic iDFT Matrix is:

$$
\frac{1}{N}
\begin{bmatrix}
e^{j\frac{2\pi}{N}\cdot 0} & e^{j\frac{2\pi}{N}\cdot 0} & e^{j\frac{2\pi}{N}\cdot 0} & \dots & e^{j\frac{2\pi}{N}\cdot 0} \\
e^{j\frac{2\pi}{N}\cdot 0} & e^{j\frac{2\pi}{N}\cdot 1} & e^{j\frac{2\pi}{N}\cdot 2} & \dots & e^{j\frac{2\pi}{N}\cdot(N-1)} \\
e^{j\frac{2\pi}{N}\cdot 0} & e^{j\frac{2\pi}{N}\cdot 2} & e^{j\frac{2\pi}{N}\cdot 3} & \dots & e^{j\frac{2\pi}{N}\cdot 2(N-1)} \\
\vdots & & & & \\
e^{j\frac{2\pi}{N}\cdot 0} & e^{j\frac{2\pi}{N}\cdot(N-1)} & e^{j\frac{2\pi}{N}\cdot 2(N-1)} & \dots & e^{j\frac{2\pi}{N}\cdot(N-1)^2}
\end{bmatrix}
$$

For both of these, down is wavenumber k and across is sample number n.
Let $\mathcal{A}$ denote the DFT matric and $\mathcal{B}$ denote the iDFT matrix.

For $N = 2$, this is:

$$
\mathcal{A} = \begin{bmatrix} 1 & 1 \\ 1 & e^{-j\pi} \end{bmatrix}
$$

$$
\mathcal{B} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & e^{j\pi} \end{bmatrix}
$$

$$
\mathcal{AB} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & e^{-j\pi} \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & e^{j\pi} \end{bmatrix} = \begin{bmatrix} 1 & \frac{1+e^{j\pi}}{2} \\ \frac{1+e^{-j\pi}}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}
$$

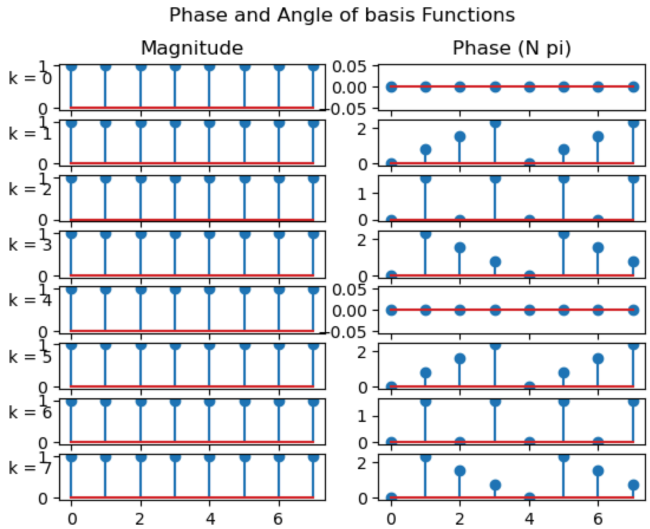
For $N = 3$, this is

$$
\mathcal{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{-j\frac{2\pi}{3}} & e^{-j\frac{4\pi}{3}} \\ 1 & e^{-j\frac{4\pi}{3}} & e^{-j\frac{8\pi}{3}} \end{bmatrix}
$$

$$
\mathcal{B} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{j\frac{2\pi}{3}} & e^{j\frac{4\pi}{3}} \\ 1 & e^{j\frac{4\pi}{3}} & e^{j\frac{8\pi}{3}} \end{bmatrix}
$$

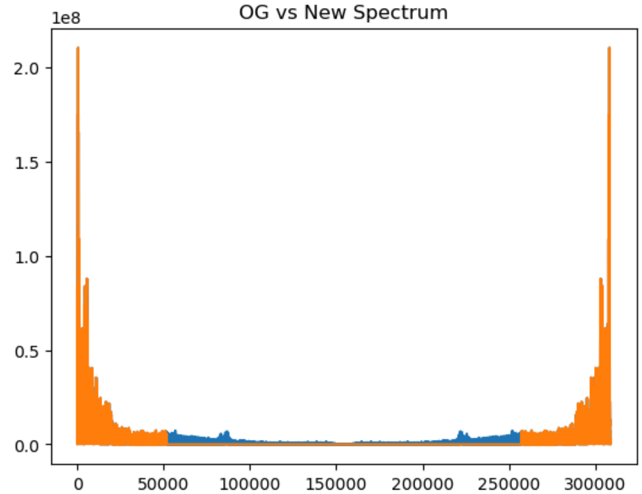
$$
\begin{aligned}
\mathcal{AB} &= \frac{1}{3}\begin{bmatrix} 3 & 1+e^{j\frac{2\pi}{3}}+e^{j\frac{4\pi}{3}} & 1+e^{j\frac{4\pi}{3}}+e^{j\frac{8\pi}{3}} \\ 1+e^{-j\frac{2\pi}{3}}+e^{-j\frac{4\pi}{3}} & 1+e^{\frac{-2\pi}{3}}e^{j\frac{2\pi}{3}}+e^{-j\frac{8\pi}{3}}e^{j\frac{8\pi}{3}} & 1+e^{-j\frac{2\pi}{3}}e^{j\frac{4\pi}{3}}+e^{-j\frac{4\pi}{3}}e^{j\frac{8\pi}{3}} \\ 1+e^{-j\frac{4\pi}{3}}+e^{-j\frac{8\pi}{3}} & 1+e^{-j\frac{4\pi}{3}}e^{\frac{2\pi}{3}}+e^{-j\frac{8\pi}{3}}e^{j\frac{4\pi}{3}} & 1+e^{\frac{-4\pi}{3}}e^{\frac{4\pi}{3}}+e^{\frac{-8\pi}{3}}e^{\frac{8\pi}{3}} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned}
$$

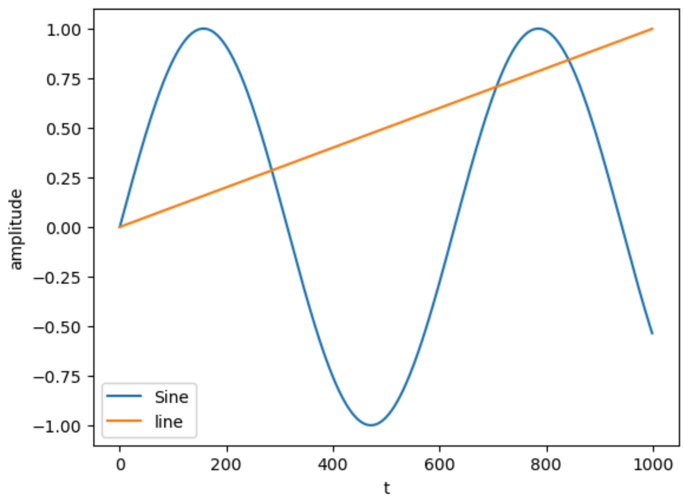
For $N = 8$, draw the basis vectors $v_k[n]$, $k = 0,1,2,\ldots,7$ as signals. Each basis vector should have a separate magnitude and phase plot. Plot phase values between 0 and 2π.

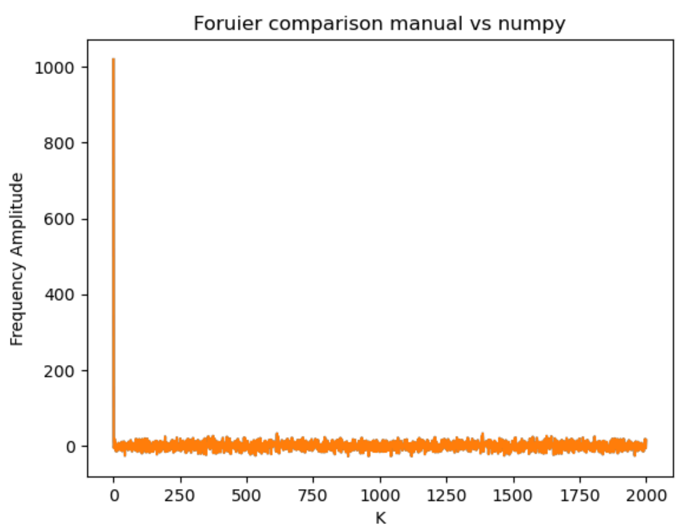


Install anaconda on your laptop and create a virtual environment for this class following the instructions on Canvas. Open the provided jupyter notebook numpyExercises.ipynb and re-implement the audio compression demo from class, making sure to set two-thirds of the FFT coefficients to zero. You may use the provided sample.wav as an audio sample. Make sure your reconstructed time-domain signal is a purely real signal. Explain how and why you set the coefficients to zero in the way that you did. Hint: The property in problem 2 above is relevant.

The Fourier series is computed in order from 0 to high frequency, and then continuing higher - at frequencies that wrap around and become negative. The two sided plot would cut this at the center and move it left. We have the highest positive and negative frequencies at the center, where there is low signal power. This data is not very useful, and can be cut.







Go through ipythonintro.ipynb, which provides a short tutorial on python, jupyter notebooks, and numpy. Then complete the exercises in numpyExercises.ipynb and submit your notebook along with a PDF containing your work for problems 0 through 4.